FAIR-Mind: Application profile for depression clinical assessments and HRV biomarker integration

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Abstract
Mental health research increasingly combines clinical assessments with physiological biomarkers, yet semantic interoperability remains limited compared to oncology and neuroimaging. FAIR-Mind is a draft Application Profile (AP) integrating depression clinical data with ECG-derived heart rate variability (HRV) biomarkers. Based on UK Biobank data structure and recent findings linking HRV to suicide risk (Weber et al., 2025), we mapped 22 variables across depression assessments (PHQ-9, CIDI, suicidality) and HRV metrics to 8 ontologies: CMO, BMONT, IOBC, BCIO, PROV-O, NCIT, CHEBI, and SNOMED CT. Ten competency questions spanning data discovery, integration, provenance, and clinical interpretation guide AP design and validation. This work demonstrates practical AP methodology for mental health research addressing calls for psychiatric ontology development (McInnis et al., 2025). The draft AP awaits UK Biobank data access (Q1 2026) for runtime validation, with community input sought on three open design questions.

Keywords
Application Profile, FAIR principles, mental health ontology, depression, heart rate variability, semantic interoperability, competency questions

1. Background

DSM and ICD classification systems lack the semantic structure needed for AI-driven analysis, contributing to the "dark data problem" in consortia like PGC and ENIGMA [1]. Ontology adoption is mature in oncology (ROO) and neuroimaging (NeuroBridge), but fragmented in mental health [2, 3]. AMP-SCZ represents the most comprehensive effort across 43 sites [4], while de novo FAIRification achieves higher compliance than retrospective approaches [5]. Depression research integrating clinical assessments with physiological biomarkers lacks comparable ontology infrastructure, motivating FAIR-Mind development.

2. FAIR-Mind Application Profile: Design and Scope

FAIR-Mind integrates depression clinical assessments with ECG-derived HRV biomarkers, motivated by Weber et al. [6] findings on suicide risk in UK Biobank. We mapped 22 variables from 500,000+ participants across four layers, composing 8 existing ontologies.

2.1. FAIR-Mind Application Profile Architecture

Three open design questions await community input: (1) suicidality granularity, (2) normalized HRV metrics representation, and (3) PHQ-9 hierarchy modeling.

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Table 1
Four-layer architecture of FAIR-Mind Application Profile

Layer	Ontologies	Data Elements	Status
Layer 1: Clinical	IOBC, SNOMED CT	PHQ-9 (items, total score, severity), CIDI, suicidality	Ontology terms identified; community input needed on suicidality granularity
Layer 2: Biomarkers	CMO, BMONT	HRV time-domain (RMSSD, SDNN), frequency-domain (LF, HF, total power), Poincaré (SD1, SD2)	Ontology terms identified; requires extension for normalized metrics and Poincaré indices
Layer 3: Provenance	PROV-O, BCIO, NCIT	ECG timestamps, device specs, processing protocols, demographics (age, sex, BMI), derivations	Ontology terms identified
Layer 4: Medications	CHEBI, BCIO	Antidepressants, lithium, mood stabilizers	Ontology terms identified

3. Community-Based Refinement Through Competency Questions

We employ competency question methodology to define AP scope and capabilities. Each question represents a realistic research query, serving as both design requirements and validation criteria.

3.1. Competency Questions by Category

Category 1: Data Discovery

- CQ1: Retrieve all participants with PHQ-9 total score >10 AND at least one HRV measurement.
- CQ2: Find all ECG recordings taken within 30 days of a depression assessment.
- CQ3: Identify participants on antidepressants WITHOUT lithium co-prescription.

Category 2: Data Integration and Comparison

- CQ4: Compare RMSSD distributions across PHQ-9 severity categories.
- CQ5: Retrieve HF power values adjusted for age and BMI (z-scored).
- CQ6: Find all HRV metrics derived from the same ECG recording session.

Category 3: Provenance and Quality Assurance

- CQ7: Identify which ECG device and processing protocol was used for each HRV measurement.
- CQ8: List all participants excluded due to cardiac arrhythmia diagnosis.
- CQ9: Retrieve the standardization method applied to age and BMI variables.

Category 4: Clinical Interpretation

- CQ10: Find participants with suicidality AND reduced parasympathetic activity (HF power <10th percentile).

3.2. Validation Approach

Phase 1 (Design-time): Expert review through community feedback (SWAT4LS 2026) on ontology selection and three open design questions. **Phase 2 (Runtime, Q1 2026):** Translation of all 10 competency questions into SPARQL queries against UK Biobank data. **Success criteria:** All questions answerable with $\leq$20 triple patterns each.

4. Expected Outcomes and Impact

Immediate Deliverables (Q1 2026). Upon UK Biobank data access: (1) Validated FAIR-Mind AP v2.0 with all 10 competency questions answered; (2) RDF schema in Turtle format with SHACL constraints; (3) Annotated synthetic dataset preserving statistical properties.

Methodological Contributions. This work demonstrates practical AP development for domains lacking mature ontology infrastructure. The competency question-driven approach with explicit documentation of open design questions provides a template for community-engaged ontology selection. The two-phase validation offers a systematic evaluation framework applicable to other AP development efforts.

Long-term Impact. FAIR-Mind will serve as a replication package for depression and biomarker integration studies, reducing barriers to FAIR adoption in mental health research. Methodology documentation will guide researchers in other psychiatric domains. Planned demonstration comparing ML model performance on raw versus AP-annotated data will test whether ontology annotation improves interpretability and enables cross-study transfer learning, addressing McInnis et al.'s (2025) call to transform "dark data" into actionable insights.

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Declaration on Generative AI

During the preparation of this work, the author(s) used Notion AI (Claude) in order to: Grammar and spelling check, Paraphrase and reword, and improve clarity of writing. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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